

Dwight Meglan, Ph.D.

Medical Device Technology Developer & Creator · **Surgical Robotics, Simulation & Image-Guided Systems**

FOUNDER · CHIEF TECHNOLOGY OFFICER · INVENTOR · TECHNICAL DILIGENCE & EXPERT WITNESS

Needham, MA

dmevlan@gmail.com

[linkedin.com/in/dmevlan](https://www.linkedin.com/in/dmevlan)

35+

YEARS BUILDING
MEDICAL DEVICES & SYSTEMS

15

COMMERCIAL SURGICAL
ROBOT PROGRAMS

90+

U.S. & INTERNATIONAL
PATENT FAMILIES

>\$15M

RESEARCH GRANTS
AUTHORED & WON

6+

VENTURES FOUNDED
MULTIPLE EXITS



CTO · surgical simulation



CTO · VIST simulator



HT Medical · medical VR & haptics



musculoskeletal simulation



Team 1757 lead mentor · 2008-present

**SIMULATION &
TRAINING**



Hugo · 3D-viz lead / fellow



robotic surgery



first NOTES robot



epicardial robot · founder



lung therapy · CTO



Ottava robotics



co-founder · architect



stroke robotics



CSP pacing guidance · founder & CTO



modular colonoscope · founder & CTO



consultancy · development · expert witness

MEDICAL ROBOTICS & DEVICES



iPlatform / Monarch



Spirus spiral endoscope · \$60M exit



insulin patch pump



endovascular robot · CTO



cardiac stimulation · CTO



Director, Medical Applications · MERL



orthopedic biomechanics fellow



Ph.D. · gait analysis laboratory



ASSERT Centre · visiting professor



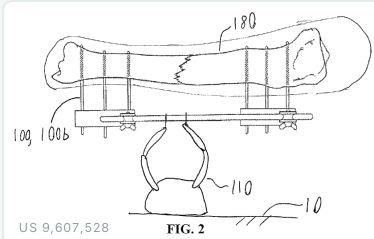
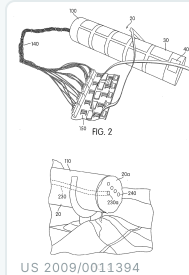
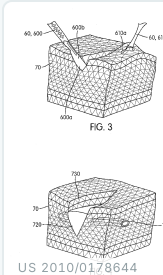
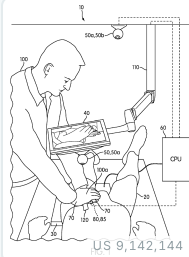
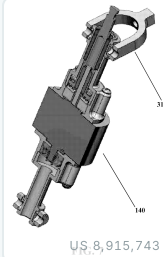
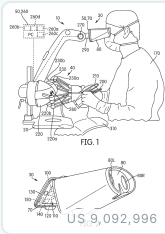
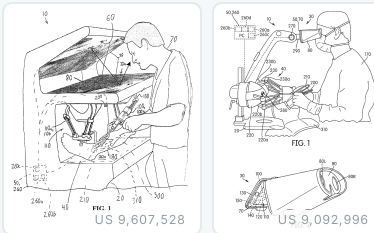
HeartLander origin license



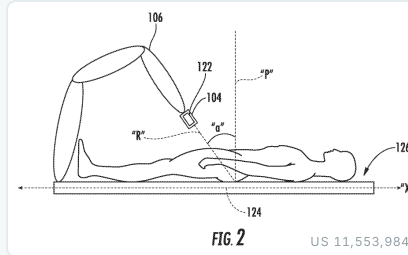
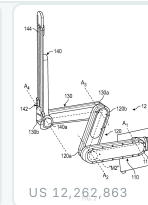
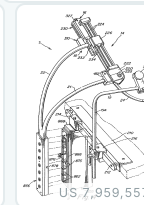
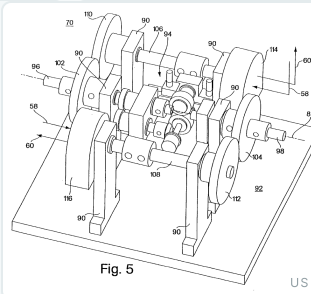
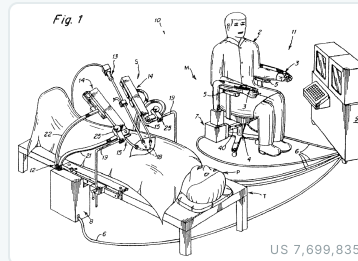
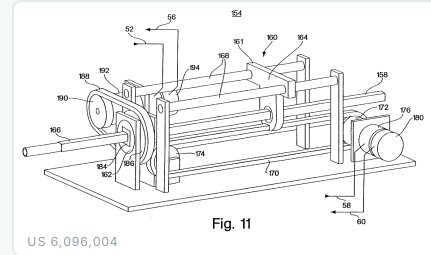
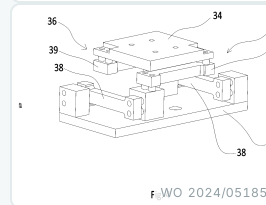
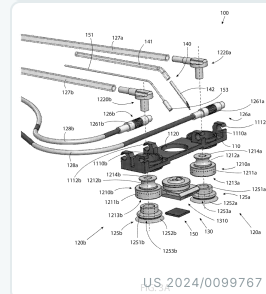
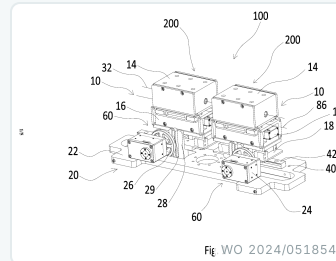
biomedical business development

**RESEARCH & INDUSTRY
LABS**

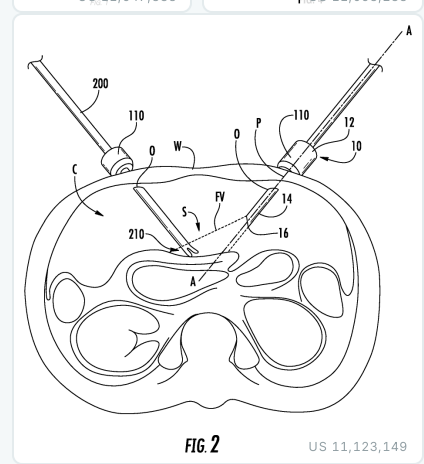
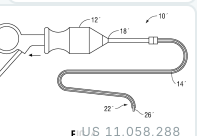
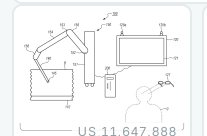
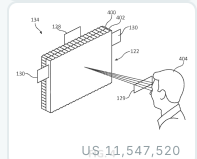
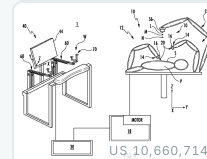
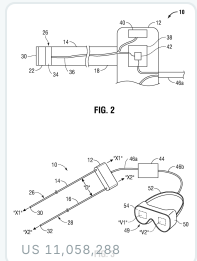
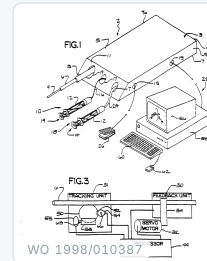
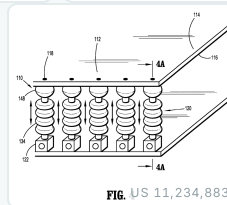
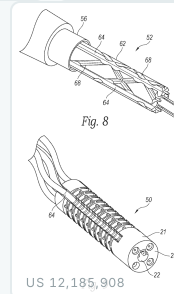
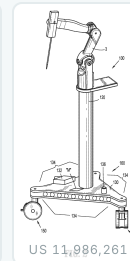
Education The Ohio State University: Ph.D. Mechanical Engineering (1991) · M.Sc. Biomedical Engineering (1985) · B.Sc. Mechanical Engineering (1983)
Postdoctoral Research Fellow, Orthopedic Biomechanics, The Mayo Clinic (1991–1993)



SIMULATION & TRAINING



MEDICAL ROBOTICS & DEVICES



VISUALIZATION, SENSING & GUIDANCE

All drawings are from granted U.S. and international patents on which Dr. Meglan is an inventor: public records; each figure links to its patent on Google Patents.

03 Medical Device & Robotics Experience: Role & Contribution Matrix

DWIGHT MEGLAN, PH.D.

PERIOD	ORGANIZATION	ROLE	PROJECT / FOCUS	ARCHITECT	FOUNDER / CTO	DETAILED DESIGN	BUILD & TEST	PRODUCT / TRIAL	IP
2023-pres	Cardathea •	Founder & CTO	CSP pacing-lead placement guidance	✓	✓	✓	✓	✓	✓
2022-pres	Rocky Mountain Biphasic •	CTO	Novel cardiac / insulin stimulation	✓	✓	✓	✓	✓	✓
2022-pres	Elements Endoscopy •	Founder & CTO	Modular drive-by-wire colonoscope	✓	✓	✓	✓	✓	✓
2022-pres	Isola Therapeutics •	CTO	Aerosolized / fluidic lung therapy	✓	✓	✓	✓	✓	✓
2021-pres	Angio8 •	Founder & CTO	Endovascular catheterization robot	✓	✓	✓	✓		✓
2017-pres	Exemplar Devices •	Principal / Consultant	Device development • diligence • expert witness	✓		✓	✓		✓
2019-2023	Auris / J&J Robotics	Consultant	Ottava robot: full-arm contact sensorization	✓		✓	✓		✓
2013-2017	Medtronic (Covidien)	System Engineer Fellow	Hugo robot: 3D-visualization lead	✓	✓	✓	✓	✓	✓
2013-2019	Univ. College Cork	Visiting Research Prof.	Surgical-training research (ASSERT)						
2005-2025	HeartLander Surgical	Founder & CTO	Epicardial VT-therapy robot (NIH SBIR) • Cardathea predecessor	✓	✓	✓	✓	✓	✓
2010-2011	Spirus → Olympus	Independent Designer	Powered spiral-endoscope electronics	✓		✓	✓	✓	
2004-2013	SimQuest International	CTO	Surgical simulation (>\$14M in grants)	✓	✓	✓	✓	✓	✓
2003-2004	Foster-Miller	Biomed Bus. Dev. Lead	Medical-device practice expansion						
2002-2003	Hansen Medical	Chief System Architect	Telerobotic catheterization (co-founder)	✓	✓				✓
2001-2002	endoVia Medical	Sr. Program Manager	First endoluminal NOTES robot	✓	✓	✓	✓	✓	✓
1999-2001	Mentice	CTO	VIST interventional simulator	✓	✓	✓	✓	✓	✓
1996-1999	Mitsubishi (MERL)	Director, Medical Apps	Interventional-cardiology simulator	✓	✓	✓	✓	✓	✓
1994-1996	HT Medical → Immersion	Engineering Coordinator	Virtual surgical environments (VR)	✓	✓	✓	✓		✓
1993-1994	MusculoGraphics	Sr. Research Scientist	Musculoskeletal-simulation software	✓		✓	✓		
1991-1993	Mayo Clinic	Research Fellow	Orthopedic biomechanics	✓		✓	✓		
1989-1991	Ohio State University	Research Engineer	Gait-analysis laboratory			✓	✓		

• = active engagement. ✓ = direct, hands-on contribution in that capacity. Additional consulting / advisory: Neptune Medical (GI robotics) • Imperative Care (stroke robotics) • Thermalin (insulin patch pump) • Dyme Medical (insulin delivery) • MIIS (imaging).

Inventor on **90+ U.S. and international patent and application families** spanning surgical robotics, simulation, endoscopy, imaging, and interventional cardiology; assignees include Covidien LP (Medtronic Surgical Innovations), Cilag (J&J), Hansen Medical, Mitsubishi Electric, Immersion, SimQuest, Elements Endoscopy, and HeartLander. Foundational grants include **US 6,096,004** (master/slave control of tubular medical tools, the basis of Hansen Medical) and **US 11,547,520** (robotic device & viewer-adaptive stereoscopic display). Patent numbers below link to the full records.

PATENT	TITLE	ASSIGNEE	PATENT	TITLE	ASSIGNEE
US 12,567,506	Simulation-based surgical procedure planning	Cilag (J&J)	US 11,547,520	Robotic device & stereoscopic display	Covidien LP
US 12,491,028	Suturing guidance system	Covidien LP	US 11,529,038	Endoscope with IMU / haptics	Elements Endoscopy
WO 2025/226661	Endoscope handle sterilizer	Elements Endoscopy	US 11,258,964	Camera-viewpoint synthesis in MIS	Covidien LP
US 12,440,286	Partial-reusable endo-tool robot	Angio8	US 11,234,883	Operating table for robotic systems	Covidien LP
US 12,357,406	Robotic surgical collision detection	Covidien LP	US 11,123,149	Angled-endoscope visualization	Covidien LP
US 12,272,463	Methods for surgical simulation	Cilag (J&J)	WO 1998/010387	Interventional-radiology interface	Immersion
US 12,262,863	Image mapping & fusion during surgery	Covidien LP	US 10,834,332	Camera viewpoints in MIS	Covidien LP
US 12,185,908	Endoscope with IMU / haptic controls	Elements Endoscopy	US 10,660,714	Optical force sensor	Covidien LP
US 12,106,508	Mitigating collision of a robotic system	Covidien LP	US 9,607,528	Combined tissue & bone simulator	SimQuest
US 12,102,288	Multi-use endoscopes	Elements Endoscopy	US 9,142,144	Hemorrhage-control simulator	SimQuest
US 11,986,261	Surgical robotic-cart placement	Covidien LP	US 9,092,996	Microsurgery simulator	SimQuest
WO 2024/051854	Catheter robot with force measurement	Angio8	US 8,915,743	Burr-hole drilling simulator	SimQuest
US 2024/0099767	Medical diagnosis & treatment system	HeartLander	US 8,671,950	Robotic medical instrument system	Hansen Medical
US 11,583,356	Controller for imaging device	Covidien LP	US 7,699,835	Robotically controlled instruments	Hansen Medical
US 11,548,140	Radio-based arm-cart location	Covidien LP	US 6,096,004	Master/slave tubular medical tools	Mitsubishi MERL

FURTHER GRANTED & PUBLISHED FAMILIES: A SELECTION

US 12,453,548 · US 12,310,687 · US 12,256,890 · US 12,193,765 · US 12,029,517 · US 12,004,773 · US 11,576,739 · US 11,517,183 · US 10,925,636 · US 10,779,856 · US 10,251,672 · EP 3,359,075 · US 11,553,984 · US 11,058,288 · US 7,959,557 · WO 2021/242681 · WO 2021/133483 · WO 2021/067591 · WO 2019/204013 · WO 2019/204011 · US 2020/0184638 · US 2019/0088162 · US 2018/0310875 · US 2010/0178644 · US 2009/0011394 · US 2007/0123748

Plus 40+ additional U.S., European, Japanese, Chinese, and PCT filings across all domains; full list available on request, or via [Google Patents](#).

GRANTS & CONTRACTS AUTHORED: >\$15M AWARDED

Precise Minimally Invasive Treatment of Ventricular Tachycardia	NIH Fast-Track SBIR
Modular Flexible Semi-Disposable Endoscope Platform	NIH SBIR I
Drug Delivery by Endoesophageal Lavage for Esophageal Cancer	NCI SBIR I
HeartLander as an Epicardial Injection-Delivery System	NIH SBIR I
Open-Source Surgical Simulation for Combat Casualty Care	U.S. Army
Intracranial Hematoma / Burr-Hole & Trauma-Flap Simulator	Army SBIR I / II / III
AR System to Teach Hemostatic-Agent Use	NIH SBIR I
Simulation-Based Wound-Closure Training System	NIH SBIR Fast-Track
Haptics-Optional Surgical Training System	Army SBIR I & II
Simulation-Based Open-Surgery Training System	Army SBIR I & II
Simulation-Based Dental Training System	NIH SBIR I & II
Leg-Trauma Wound-Simulation Trainer for Army Medics	DARPA

RECOGNITION & SERVICE

- Senior Member, IEEE
- Senior Member, ACM
- NIH Study-Section Reviewer
- U.S. Defense SBIR Reviewer
- Canadian NSERC Reviewer
- ACS Steering Committee on Surgical Simulation
- FIRST Robotics 1757 Lead Mentor: NE District Champions 2023 · Worlds top 4%

SIGNATURE ACHIEVEMENTS

- **\$60M Olympus exit:** sole developer of the UI & motor controls for Spirus Medical's powered spiral endoscope.
- **Medtronic Hugo:** 3D-visualization lead & technical fellow; low-latency stereoscopic imaging to the da Vinci standard.
- **First NOTES surgical robot:** built the first endoluminal master-slave telerobot; assets passed to Hansen & Intuitive.
- **Mentice VIST:** designer / lead engineer of the most successful endovascular-simulation product; 400+ units, 20+ years on market.
- **Co-founder, Hansen Medical:** recruited by Intuitive founder Fred Moll; foundational catheter-robot IP (US 6,096,004).
- **Medtronic robotics ML group:** identified & advocated the acquisition (Digital Surgery).
- **OpenSurgSim:** conceived & won funding for the DoD open-source surgical-simulation platform.
- **Technical & IP due diligence:** advisor to investors and acquirers of surgical-robotics and simulation companies.
- **Expert witness:** medical-device & simulation litigation; patent & trade-secret assessment, reports, depositions.

SELECTED PUBLICATIONS

- Coles T, Meglan D, John N. "The Role of Haptics in Medical Training Simulators." *IEEE Trans. Haptics*, 2011;4(1):51-66.
- Meglan D, et al. "Development of an Open Surgery Simulator." *Medicine Meets VR*, 2011.
- Champion HR, Meglan DA. "Minimizing surgical error by incorporating objective assessment." *J. Am. Coll. Surg.*, 2008.
- Dawson SL, et al. "Computer-based simulator for interventional-cardiology training." *Catheter. Cardiovasc. Interv.*, 2000.